Deep Spatial-Spectral Global Reasoning Network for Hyperspectral Image Denoising

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***Abstract*— Although deep neural networks (DNNs) have been widely applied to hyperspectral image (HSI) denoising, most DNN-based HSI denoising methods are designed by stacking convolution layer, which can only model and reason local relations, and thus ignore the global contextual information. To address this issue, we propose a deep spatial-spectral global reasoning network to consider both the local and global information for HSI noise removal. Specifically, two novel modules are proposed to model and reason global relational information. The first one aims to model global spatial relations between pixels in feature maps, and the second one models the global relations across the channels. Compared to traditional convolution operations, the two proposed modules enable the network to extract representations from new dimensions. For the HSI denoising task, the two modules, as well as the densely connected structures, are embedded into the U-Net architecture. Thus, the new-designed global reasoning network can help tackle complex noise by exploiting multiple representations, e.g., hierarchical local feature, global spatial coherence, cross-channel correlation, and multi-scale abstract representation. Experiments on both synthetic and real HSI data demonstrate that our proposed network can obtain comparable or even better denoising results than other state-of-the-art methods.**

***Index Terms*— Deep neural network (DNN), global channel module (GCM), global spatial module (GSM), hyperspectral image (HSI) denoising.**

I. Introduction

IN THE remote sensing community, it is a crucial task to remove noise from hyperspectral images (HSIs) since the noise can severely degrade the image quality, and thus impair the performance of subsequent HSI applications, such as classification [1]–[5], hyperspectral dimensionality reduction [6], object detection of HSI [7], and unmixing [8], [9]. To address this issue, a large number of HSI denoising methods have been proposed in the last few decades. Generally, these methods can be divided into two categories, namely model-driven-based methods and data-driven-based methods.

A. Related Work: Model-Driven-Based Methods and Data-Driven-Based Methods

The model-driven-based HSI denoising methods are designed by utilizing the prior knowledge of the HSI, such as spatial-spectral total variation [10], [11], spatial-spectral non-local mean [12], [13], spatial-spectral sparse representation [13]–[16], and low-rank prior [17]–[26]. For example, He *et al.* [11] assumed that HSIs are piecewise smooth in the spatial dimension and adopted the total variation to capture this property. Qian and Ye [13] introduced the nonlocal similarity and spectral-spatial structure of HSI into sparse representation. Zhao and Yang [15] adopted sparse coding to model the global redundancy and correlation (RAC) in the spatial domain and local RAC in the spectral domain. Peng *et al.* [14] utilized both the nonlocal similarity over space and the global correlation across the spectrum of HSIs and proposed a tensor dictionary learning (TDL) model. Chen *et al.* [18] modeled the noise with non-independent identically distributed (i.i.d.) mixture of Gaussian distributions in the low-rank matrix factorization model. Recently, non-local low-rank tensor-based methods, i.e., KBR [24], LLRT [25], and NGmeet [26] achieve state-of-the-art denoising results since they elaborately model the intrinsic property of the HSI.

Recently, the data-driven-based methods using deep learning have been applied to the HSI denoising task [27]–[33]. Specifically, Chang *et al.* [27] first introduced the deep convolutional neural network for HSI denoising. Yuan *et al.* [28] developed a spatial-spectral deep residual convolutional neural network with multi-scale and multi-level feature representation. Dong *et al.* [29] presented a modified 3-D U-net architecture to exploit spatial-spectral correlations in HSIs. Wei *et al.* [30] introduced a quasi-recurrent pooling function into the 3-D U-net to further capture the global correlation along the spectrum. Zhang *et al.* [31] presented a new deep CNN by incorporating the spatial-spectral gradient information. Liu and Lee [32] proposed a 3-D dilated CNN for HSI denoising. Zhang *et al.* [33] proposed a novel deep

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TABLE I
COMPARISONS OF DIFFERENT DEEP LEARNING-BASED HSI DENOISING METHODS

Priors/Methods	Chang et al. [27]	Yuan et al. [28]	Dong et al. [29]	Wei et al. [30]	Zhang et al. [31]	Liu et al. [32]	Zhang et al. [33]	Ours
network	ResNet	ResNet	U-Net	U-Net + RNN	Plain CNN	ResNet	Plain CNN	U-Net
local spectral information	✓	✓	✓	✓	✓	✓	✓	✓
local spatial information	✓	✓	✓	✓	✓	✓	✓	✓
global spectral information	×	×	×	✓	×	×	×	✓
global spatial information	×	×	×	×	×	×	×	✓

spatial-spectral Bayesian posterior (DSSBP) framework for non-i.i.d. noise removal. For convenience, we summarize and compare these deep learning-based HSI denoising methods in Table I.

B. Motivations and Contributions

Although these deep learning-based HSI denoising methods achieve excellent performance, they still have not sufficiently explored the local and global spatial-spectral information of the HSIs. As shown in Table I, there are four kinds of prior information that can be exploited to help improve the denoising performance of the models, namely local spectral information, local spatial information, global spectral information, and global spatial information. However, none of the existing deep learning-based methods take all these prior information into consideration. Therefore, to fill this gap and further improve the denoising performance of these deep learning-based methods, we design a global reasoning network to fully exploit all the four kinds of priors for HSI denoising.

More specifically, two global reasoning modules are proposed to extract relation-aware features. The first one is global spatial modules (GSMs), which are derived from the non-local neural network [34] and used to reason global spatial relationships between pixels in the feature map. Thus, the global spatial-spectral correlation of the HSIs can be well utilized. Also, the local spatial-spectral correlations can be well characterized by the traditional 2-D convolution. Therefore, the global and local spatial-spectral correlation of the HSIs can be fully exploited. The second one is the global channel module (GCM), which is based on graph convolution network (GCN) [35] and is used to reason global interdependencies across channels of the feature map. Thus the global correlation along the spectrum of the HSIs can be well characterized. Furthermore, combined with the densely connected module (DCM) in [36], the two modules can construct two basic units separately, which are then embedded into the classical multi-scale U-net [37] to construct the global reasoning network for HSI denoising. In this way, the new-designed network can help tackle complex noise by exploiting multiple representations, such as hierarchical local and global features, cross-channel correlation, and multi-scale abstract representation. Additionally, since the proposed two modules, e.g., GSM and GCM, are orthogonal, they thus can provide complementary information for each other, which makes the proposed network extract feature representations better, and thus helps improve the final denoising performance.

In summary, motivated by alleviating the issue of insufficient exploration for the global spatial and spectral correlation information of HSIs in the existing deep learning-based HSI denoising methods, we propose two novel global reasoning modules, e.g., GSM and GCM, and then form two basic units by combining with the DCM. Furthermore, the two new units are embedded into the multi-scale U-net to construct the global reasoning network. Compared with the existing networks for HSI denoising, the proposed network can better model and reason the global spatial and spectral correlation information and is also easy to implement. More specifically, the contributions of our proposed method are summarized as follows.

1) We propose two global reasoning modules that can effectively capture the global spatial coherence and channel correlation. By combining the two modules with a DCM, we develop two basic units, which can help extract rich relation-aware information for learning better denoising task-specific representations.
2) We propose a global reasoning network for HSI denoising by inserting the two basic units into the multi-scale U-net. The proposed network cannot only learn richer feature representations in the encoding phase but also obtain better denoising results in the decoding phase.
3) Experimental results on both synthetic and real HSI data sets illustrate that our proposed model can achieve comparable or better performance compared with other state-of-the-art methods.

The rest of this article is organized as follows: In Section II, we introduce the proposed network in detail. In Section III, we conduct a variety of experiments on synthetic and real HSI data sets. In Section IV, we draw the conclusion.

II. PROPOSED MODEL

From a generative perspective, the observed noisy HSIs $\mathbf{Y} \in \mathbb{R}^{H\times W\times B}$ where H and W are the height and width of the spatial image, respectively, and B is the number of spectral bands, can be modeled as

$$\mathbf{Y} = \mathbf{X} + \mathbf{N} \tag{1}$$

where $\mathbf{X} \in \mathbb{R}^{H\times W\times B}$ is the clean HSIs, and $\mathbf{N} \in \mathbb{R}^{H\times W\times B}$ is the addictive noise, i.e., Gaussian noise, sparse noise (e.g., stripe, deadline, impulse), or a mixture of them [19], [20]. The goal of HSI denoising is to recover the clean image $\mathbf{X}$ from the observed noisy image $\mathbf{Y}$. In this section, we propose a global reasoning network to accomplish this HSI denoising task. In the following, we will first introduce the overall network architecture, and then present our proposed core modules of the network detailedly.

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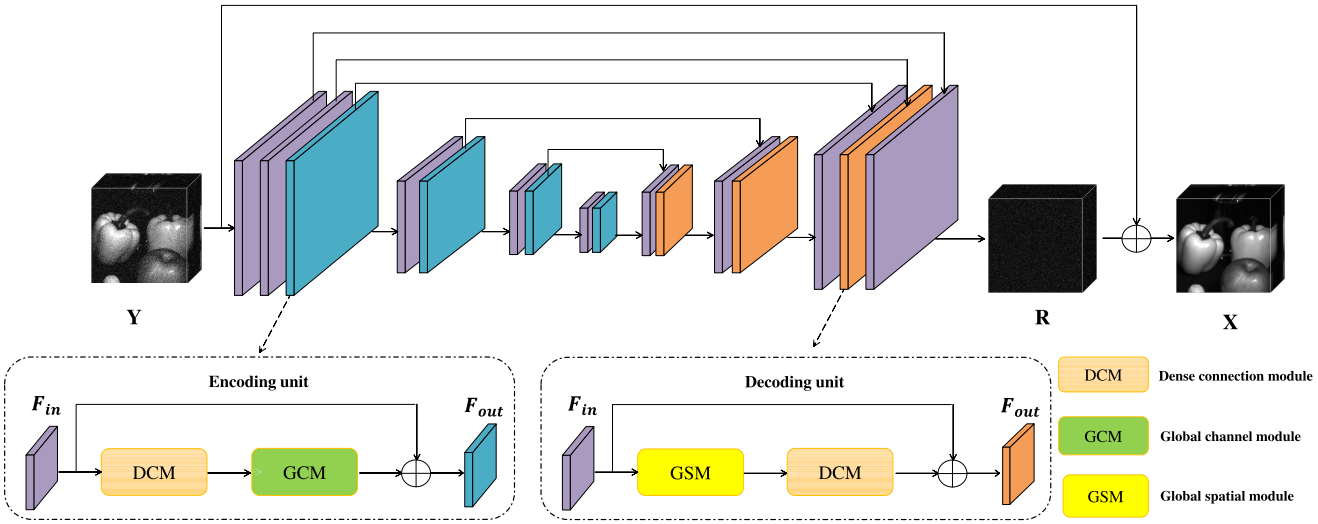

Fig. 1. Overall architecture of our global reasoning network for HSI denoising. The network consists of encoding and decoding units. The encoding unit contains the DCM and the GCM, while the decoding unit contains the GSM and the DCM.

A. Overall Network Architecture

The proposed network adopts the U-net architecture [37] as the backbone, and the advantages of the U-net are three folds: First, the skip connections in U-net allow the restored feature map in the decoding stage to contain more low-level features and promotes gradient back-propagation, which make the restored image more realistic and help network training. Second, the U-net structure is able to capture multi-scale features, which has been verified to be very effective in the natural image restoration tasks (e.g., [38], [39]), thus is very probable to be effective for the HSI denoising tasks. Last but not least, compared with other more advanced and complex deep networks, the structure of U-net networks is relatively simple, and is very easy to be implemented. The proposed network takes noisy HSIs as input and predicts its denoised version, and contains two feature extraction layers, four encoding units, three decoding units and one reconstruct layer as shown in Fig. 1. Specifically, each encoding unit is built based on the DCM and GCM and thus is able to extract abstract features and the reason the relations of these feature maps along the channel dimensions. As for the decoding unit, it is constructed based on the DCM and GSM and thus can reconstruct clean images by up-sampling features and reasoning contextual information along the spatial dimensions in the features.

The feature extraction layer is designed to extract shallow features from the noisy HSIs input. The operations of the two layers are defined as

$$\mathbf{F}_1 = \mathbf{W}_1 \otimes \mathbf{Y} \tag{2}$$

$$\mathbf{F}_2 = \mathbf{W}_2 \otimes \mathbf{F}_1 \tag{3}$$

where $\mathbf{F}_i, i = 1, 2$ are the feature maps, $\mathbf{W}_i, i = 1, 2$ are the kernels, and $\otimes$ is convolutional operation. As shown in Fig. 1, these two shallow features $\mathbf{F}_1$ and $\mathbf{F}_2$ are further propagated to deeper layers using skip-connections, which is beneficial for preserving raw information and avoiding the gradient vanishing during training. The reconstruction layer is used to generate the denoised image $\hat{\mathbf{X}}$ as follows:

$$\hat{\mathbf{X}} = \mathbf{Y} + \hat{\mathbf{N}} = \mathbf{Y} + g(\mathbf{Y}). \tag{4}$$

It should be noted that the output of the global reasoning network $g(\cdot)$ is the noise $\hat{\mathbf{N}}$, which is a commonly used residual learning strategy to help training [40].

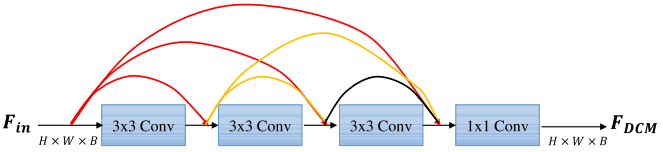

Fig. 2. Architecture of DCM.

B. Proposed Modules

In this section, we will introduce one popular existing module and two proposed modules in detail for the specific HSI denoising task, i.e., one DCM [36] and two global reasoning modules. Before introducing the proposed two global modules, we first provide a detailed description on the existing DCM.

1) Dense Connected Module: Since hierarchical features extracted by a traditional convolutional neural network (CNN) with stacked convolutional layers can capture the common characteristics of similar objects at different local areas, and help improve the image denoising performance, we thus adopt a L-layer DCM proposed in [36] to extract local dense hierarchical features as show in Fig. 2. The output of the lth layer of this DCM can be calculated as

$$\mathbf{F}_l = \sigma\left(\mathbf{W}_l \otimes \text{fuse}(\mathbf{F}_{\text{in}}, \mathbf{F}_1, \ldots, \mathbf{F}_{l-1})\right), \quad l = 1, \ldots, L \tag{5}$$

where $\sigma(\cdot)$ is the relu$(\cdot)$ activation function, $\mathbf{W}_l$ is the kernel and fuse$(\cdot)$ is the fusion operation. By using this densely connected structure, we can obtain multi-level local spatial

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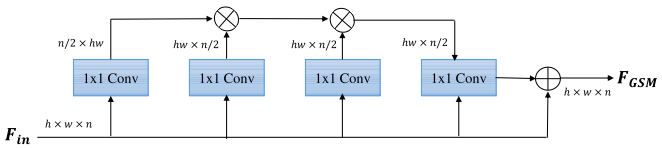

Fig. 3. Architectures of GSM.

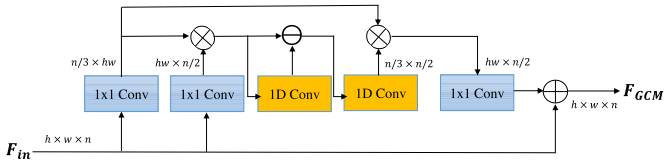

Fig. 4. Architectures of GCM.

feature representations, which can be further fused with a 1×1 convolutional operation as follows:

$$\mathbf{F}_{\mathrm{DCM}} = \mathbf{W}_{\mathrm{fuse}} \otimes \mathrm{fuse}(\mathbf{F}_{\mathrm{in}}, \mathbf{F}_1, \ldots, \mathbf{F}_L) \tag{6}$$

where $\mathbf{F}_{\mathrm{DCM}}$ is output of this module, and $\mathbf{W}_{\mathrm{fuse}}$ is the kernel of this fusion operation. To save the memory space, the fusion operation fuse$(\cdot)$ is set as the add$(\cdot)$ operation instead of the common used concat$(\cdot)$ operation. Empirically, we find the add$(\cdot)$ fusion operation works well. Although this DCM generates hierarchical feature representation with increasing receptive fields, the extracted information of the fused features still comes from a local spatial region. To alleviate this issue, we further propose two global reasoning modules to capture feature representations of global spatial coherence and global channel correlation of the HSIs.

2) Global Spatial Module: To model the global spatial coherence of the HSIs, we design a GSM as shown in Fig. 3. Specifically, this module can be used to reason the relations between one pixel and all the other pixels in the feature map.

We denote the input feature map as $\mathbf{F}_{\mathrm{in}} \in \mathbb{R}^{h \times w \times n}$, where h, w and n are the height, the width and the number of channel, respectively. Before inputed into this module, the feature map $\mathbf{F}_{\mathrm{in}}$ is first unfolded as the shape of $hw \times n$ along the spatial dimension. In this module, we adopt three 1×1 convolutional layers, i.e., $\alpha(\cdot)$, $\beta(\cdot)$ and $\gamma(\cdot)$, on the input feature map to reduce the channel number from n to $n/2$. As shown in Fig. 3, the output of the GSM can be computed as

$$\mathbf{F}_{\mathrm{GSM}} = \mathbf{F}_{\mathrm{in}} + \alpha(\mathbf{F}_{\mathrm{in}})\beta(\mathbf{F}_{\mathrm{in}})^T \gamma(\mathbf{F}_{\mathrm{in}})\mathbf{W}_{\mathrm{GSM}} \tag{7}$$

where T is the transpose operation, and $\mathbf{W}_{\mathrm{GSM}}$ is the 1×1 convolutional kernel of the output layer. It should be noted that $\alpha(\mathbf{F}_{\mathrm{in}})\beta(\mathbf{F}_{\mathrm{in}})^T$ can be regarded as calculating the relationships of pixels, and the output dimension is $hw \times hw$. In other words, $\alpha(\mathbf{F}_{\mathrm{in}})\beta(\mathbf{F}_{\mathrm{in}})^T$ can diffuse information across pixels. Therefore, the function of $\gamma(\mathbf{F}_{\mathrm{in}})$ is to aggregate information from pixels, and $\alpha(\mathbf{F}_{\mathrm{in}})\beta(\mathbf{F}_{\mathrm{in}})^T$ measures the similarity between these pixels, i.e., depicting the spatial relationship among pixels. To further make this clear, we show one feature map learned by the GSM in Fig. 5(c), from which we can easily observe that the features from the GSM have large response to similar or duplicate information in long-range context, e.g., the two car's edges, which proves its effectiveness in exploring the spatial relationship among pixels. Additionally, we reorder the computation of $\alpha(\mathbf{F}_{\mathrm{in}})\beta(\mathbf{F}_{\mathrm{in}})^T\gamma(\mathbf{F}_{\mathrm{in}})$ as $\alpha(\mathbf{F}_{\mathrm{in}})(\beta(\mathbf{F}_{\mathrm{in}})^T\gamma(\mathbf{F}_{\mathrm{in}}))$ by utilizing the associative rule, which can dramatically reduce the computational complexity compared with the non-local module [36]. In summary, this GSM can make the network generate relational predictions that take all pixels of the feature map into account, which is very beneficial to extract information about object structures.

3) Global Channel Module: Furthermore, to model the channel correlations of the feature map, we propose a GCM based on the graph convolutional network [35], which can be used to model long-range information and is shown in Fig. 4. Specifically, the output of the GCM is defined as

$$\mathbf{F}_{\mathrm{GCM}} = \mathbf{F}_{\mathrm{in}} + \mathbf{A}_{\mathrm{GCM}}\mathbf{F}_c\mathbf{W}_{\mathrm{GCM}} \tag{8}$$

where $\mathbf{A}_{\mathrm{GCM}}$ is the adjacent matrix of the graph, $\mathbf{F}_c$ is the feature matrix, and $\mathbf{W}_{\mathrm{GCM}}$ is the weight matrix. The feature matrix $\mathbf{F}_c$ can be computed as

$$\mathbf{F}_c = \mu(\mathbf{F}_{\mathrm{in}})^T \tau(\mathbf{F}_{\mathrm{in}}) \tag{9}$$

where $\mu(\mathbf{F}_{\mathrm{in}})\mathbb{R}^{hw \times \frac{n}{3}}$ and $\tau(\mathbf{F}_{\mathrm{in}}) \in \mathbb{R}^{hw \times \frac{n}{2}}$ are two 1×1 convolutional operations on the input feature map $\mathbf{F}_{\mathrm{in}}$, and $\mathbf{F}_c \in \mathbb{R}^{\frac{n}{3} \times \frac{n}{2}}$ is able to capture the channel correlation information. For the new feature matrix $\mathbf{F}_c$, we can build a new graph, where $(n/3)$ is the number of nodes, and $(n/2)$ is the feature dimension of each node. Therefore, the output of this module is computed as

$$\begin{aligned} \hat{\mathbf{F}}_c &= (\mathbf{I} + \mathbf{A}_{\mathrm{GCM}})\mathbf{F}_c\mathbf{W}_{\mathrm{GCM}} \\ &= (\mathbf{I} + \mathbf{A}_{\mathrm{GCM}})(\mu(\mathbf{F}_{\mathrm{in}})^T\tau(\mathbf{F}_{\mathrm{in}}))\mathbf{W}_{\mathrm{GCM}} \end{aligned} \tag{10}$$

where the adjacent matrix $\mathbf{A}_{\mathrm{GCM}}$ and the weight matrix $\mathbf{W}_{\mathrm{GCM}}$ are implemented using the 1-D convolution. Here, to add the numerical instabilities, we add an identity matrix $\mathbf{I}$ into the adjacent matrix $\mathbf{A}_{\mathrm{GCM}}$ as has been done in [35]. To make the output of this module be used in the subsequent operations, two additional operations are utilized. First, we multiply $\mu(\mathbf{F}_{\mathrm{in}})$ by the obtained feature matrix $\hat{\mathbf{F}}_c$ and then use a 1×1 convolutional layer to increase the number of channel to n. By embedding the proposed GCM into the basic unit, our GCM models interdependencies along the channel dimensions of the network's feature map, and captures correlations between more abstract features in the image, i.e., the spectral relationship. This is because the calculation process of $\mathbf{F}_c$, i.e., (9), is to directly pixel-wise multiply the two feature maps and then add them. In this way, spectral information from the two feature maps can be aggregated and pairwise interactions can be modeled. Furthermore, each node contained in the adjacency matrix $\mathbf{A}_{\mathrm{GCM}}$ can be seen as the spectra descriptor, and can be determined by learning. Therefore, our GCM is able to describe the spectral interrelationship along the channel dimension. To make this clear, we show one feature map learned by the GCM in Fig. 5(b), from which it can be easily seen that the features from the GCM have large response to content, i.e., the whole sky, which has a close

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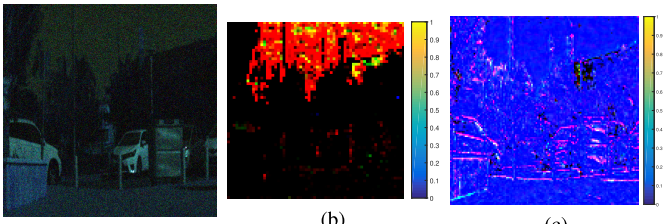

(a) (b) (c)

Fig. 5. Visualizations of the two global modules. (a) Input noisy image. (b) Feature map learned by the GCM. (c) Feature map learned by the GSM.

relationship with spectral information. This similar with high-level vision tasks that capture semantic features, and proves the effectiveness of GCM in exploring the spectral relationship among feature maps.

C. Proposed Global Reasoning Network

In this section, we provide a summary of our proposed global reasoning network. In general, two basic units are further constructed based on the popular DCM and two proposed modules above, and then embedded into the U-net to build our final global reasoning network for HSI noise removal. More specifically, the DCM and the GCM are cascaded together to create the encoding unit due to the high-level abstract feature representations goal of the encoder. As shown in Fig. 1, the basic encoding unit is defined as

$$\mathbf{F}_{\text{eu}} = \mathbf{F}_{\text{in}} + f_{\text{GCM}}(f_{\text{DCM}}(\mathbf{F}_{\text{in}})) \tag{11}$$

where $\mathbf{F}_{\text{eu}}$ is the output of the encoding unit, f_{GCM} and f_{DCM} represent DCM and GCM, respectively. The DCM is able to extract local dense features, while the GCM can reason the correlation between the channels of the local dense features, and thus helps capture better global information.

Additionally, we build a decoding unit by cascading the GSM and the DCM. As shown in Fig. 1, the basic decoding unit is defined as

$$\mathbf{F}_{\text{du}} = \mathbf{F}_{\text{in}} + f_{\text{DCM}}(f_{\text{GSM}}(\mathbf{F}_{\text{in}})) \tag{12}$$

where $\mathbf{F}_{\text{du}}$ is the output of the decoding unit, and f_{GSM} represents the GSM. Unlike the encoding unit, the GSM is put before the DCM, which provides global spatial information for the subsequent DCM, and thus helps to reconstruct spatially coherent structures of the HSIs.

In summary, as shown in Fig. 1, four encoding units (green cubes) and three decoding units (purple cubes) are used to build the encoder and decoder, respectively. Besides, we use strided convolution and nearest-neighbor interpolation to implement the down-sampling and up-sampling process, respectively. As aforementioned, we also utilize the skip-connection structure to propagate $\mathbf{F}_{\text{eu}}$ to the decoder, which cannot only preserve more detailed raw information and avoid the gradient vanishing, but also take full advantage of the two orthogonal global contextual information to improve the denoising performance.

D. Implementation Details

After building the denoising network, we first define the loss function of this network. As we all know that, the most widely used loss function for training a network is l_2 loss, but this loss usually generates over-smoothed results. To alleviate this issue, we adopt the l_1 loss as the final loss function in order to make a balance between noise removal and detail preservation. Specifically, this l_1 loss function is defined as

$$\mathcal{L} = \frac{1}{T}\sum_{i=1}^{T} ||\mathbf{X}_i - \mathbf{X}_i^{(gt)}||_1 \tag{13}$$

where T is the number of training patch image, $\mathbf{X}_i$ and $\mathbf{X}_i^{(gt)}$ represent the output denoised image patch and ground truth image patch, respectively.

Additionally, the implementation details are shown as follows. The size of each kernel is shown in Figs. 2–4 detailedly. The number of feature maps is set as 64 for all the convolutions. We adopt the Adam algorithm to train this network. The batch size is set as 16. The training images are cropped into 64×64 patch pairs. In the training, the learning rate is fixed as 0.0001 and the number of training epoch is set as 100.

III. Experiments

In this section, to evaluate the performance of our proposed network, we conducted a series of experiments on both synthetic and real HSI data. For the competing methods, state-of-the-art traditional and deep learning-based HSI denoising methods are both considered. Specifically, in the Gaussian noise case, eight methods are compared, including filtering-based approaches (BM4D [12]), low-rank matrix-based method (LRMR [17], LRTV [11]), dictionary learning-based approach (TDL [14]), tensor-based approaches (LLRT [25], NGmeet [26]), and deep learning-based methods (HSI-DeNet [27], HSID-CNN [28]). In the complex noise case, the competing methods include low-rank matrix-based method (LRMR [17], LRTV [11], E3DTV [41]), complex noise modeling methods (NMoG [18]), low-rank tensor-based method (LRTDTV [22]), dictionary learning-based approach (TDL [14]), tensor-based approaches (NGmeet [26]), and the aforementioned two deep learning methods (HSI-DeNet [27], HSID-CNN [28]). All the experiments are conducted on a PC with Intel Core i7-8700K CPU, NVIDIA GTX 1080Ti GPU, and 32-GB memory. Code will be available at https://github.com/xiangyongcao/GRN.

A. Synthetic Experiments

This experiment aims to evaluate the performance of the proposed network with other competing methods quantitatively. One clean HSI data set, called ICVL,[1] is used in all the synthetic experiments. Specifically, this data set contains 201 images, of which each is with a size of $1392 \times 1300 \times 31$. Some sample images of this data set are shown in Fig. 6. In our experiment, we used only 141 images in total, where 100 images are used for training, one image is for validating,

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Fig. 6. Some sample images of the ICVL data set.

and the remaining 40 images are for testing. The training images are cropped into $64 \times 64 \times 31$ patch pairs. Before conducting experiments, each image is normalized into [0, 1]. All the images in this data set can be seen as clean images.

Since real HSIs are always polluted by complex noise, such as Gaussian noise, sparse noise (stripe, deadline, or impluse), or a mixture of them, we thus add five kinds of complex noise into the clean images to simulate these real noise cases, which are shown as follows:

Case 1: Non-i.i.d. Gaussian Noise. Entries in all bands are contaminated by zero-mean Gaussian noise with different intensities, which are randomly chosen from 30 to 70.

Case 2: Gaussian + Stripe Noise. As mentioned in case 1, each band is corrupted by non-i.i.d Gaussian noise. Besides, ten bands in the ICVL data set are randomly selected to add stripe noise, and the number of stripes in each band is 5% to 15% of columns.

Case 3: Gaussian + Deadline Noise. Each band is contaminated by non.i.i.d Gaussian noise as mentioned in case 1. Then, ten bands in the ICVL data set are chosen randomly to add deadline noise. The number of deadlines in each band is 5% to 15% of columns.

Case 4: Gaussian + Impluse Noise. All bands are corrupted by Gaussian noise as mentioned in case 1. Then, ten bands in ICVL data set are randomly selected to add impulse noise with different intensities, and the percentage of impluse ranges from 10% to 70%.

Case 5: Mixture Noise. First, all bands are corrupted by Gaussian noise as previously mentioned. Then each band is randomly corrupted by at least one kind of the other four noise mentioned above.

For this synthetic experiment, we only need to train one network for all the testing cases. Specifically, the training process contains three stages. In the first stage, the zero-mean Gaussian noise with noise level $\sigma = 50$ is added into the training images to construct the training data set. In the second stage, the zero-mean Gaussian noises with different noise levels σ (uniformly selected from 30 to 70) are used to build the training data set. In the final stage, we use the aforementioned complex noises (from case 1 to case 5) to generate the training data set. As for the testing, we consider two scenarios using the 40 testing images. The first scenario aims to validate the performance of the proposed method in the Gaussian noise case with different noise levels ($\sigma = 30, 50, 70$ and blind). For each noise level, the testing data set is generated by adding the Gaussian noise with the same noise intensity to each band. The second scenario is designed to evaluate the performance of the proposed method in the complex noise cases (from case 1 to case 5). Here, it should be noted that the 40 testing images are cropped into $512 \times 512 \times 31$ for testing since some other traditional competing methods (e.g., LLRT, NGMeet) cannot afford the computational time.

Three quantitative metrics are employed for performance evaluation, including peak signal-to-noise ratio (PSNR), structure similarity (SSIM [42]), and spectral angle mapper (SAM) [43]. PSNR and SSIM are two conventional spatial-based indexes, while SAM is a spectral index. For PSNR and SSIM, a larger value indicates better performance, while a smaller value implies better performance for SAM. We also report the average running time[2] of all the cases for each method.

First, we evaluate the performance of our proposed method in the Gaussian noise case by using the trained network after stage 2. The denoised quantitative results are shown in Table II, and one denoised visual comparison result is illustrated in Fig. 7. From Table II, it can be easily seen that our method dramatically outperforms some state-of-the-art denoising methods, such as BM4D, LRMR, and LRTV. Also, as has been universally acknowledged that the LLRT and NGmeet methods are three leading methods for the Gaussian noise removal of HSIs, we can observe that our method can obtain comparable or even better denoising results in terms of the three metrics (PSNR, SSIM, and SAM) in all the Gaussian noise cases in comparison with them. This is because our method fully considers the global spectral correlation information and global spatial coherence of the HSIs as has been done by these methods. Additionally, compared with the two deep learning-based methods (HSI-DeNet and HSID-CNN), our method obviously achieves better performance since the proposed global modules in our method can better take advantage of the global spatial-spectral information of the HSIs. Also, the running time of our method is far less than the traditional methods once the network has been trained. From Fig. 7, it can be easily observed that our method can better remove the noise and preserve the detailed information. In addition, we show the PSNR value of each band for Fig. 7 in Fig. 11(a), from which it can be observed that our method can obtain higher PSNR values across almost all the bands compared with other methods.

Then, we evaluate our method in the complex noise case by using the final trained network after stage 3. The denoised quantitative results are shown in Table III, and three denoised visual results are shown from Figs. 8–10. From Table III, we can see that our method achieves better denoised results obviously compared with some state-of-the-art methods, such as LRMR, LRTV, NMoG, and E3DTV, which is because these methods are all low-rank matrix-based methods and lose some structure information in the process of denoising. Therefore, by utilizing the low-rank tensor-based methods that better characterize the HSI structure as well as the noise characteristic of HSI, LRTDTV can obtain better denoised results. Besides, it should be noted that our proposed deep learning-based method only has a marginal performance gain

[2]For the deep learning methods, the running time refers to the test time.

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TABLE II

PERFORMANCE COMPARISON OF ALL THE COMPETING METHODS IN THE GAUSSIAN NOISE CASE ON THE ICVL DATA SET. THE BEST AND THE SECOND BEST RESULTS ARE BOLDFACED AND UNDERLINED, RESPECTIVELY

	Noisy HSI	BM4D [12]	LRMR [17]	LRTV [11]	TDL [14]	LLRT [25]	NGmeet [26]	HSI-DeNet [27]	HSID-CNN [28]	Ours
						σ=30				
PSNR	18.58	38.99	33.01	32.08	41.11	42.39	**42.96**	41.92	41.95	42.69
SSIM	0.093	0.941	0.723	0.869	0.962	0.970	0.977	0.979	0.980	**0.988**
SAM	0.864	0.122	0.276	0.089	0.069	0.059	**0.048**	0.094	0.098	0.068
						σ=50				
PSNR	14.15	36.22	29.18	31.88	38.79	38.53	40.74	40.35	40.54	**40.89**
SSIM	0.037	0.902	0.554	0.867	0.941	0.925	0.962	0.968	0.973	**0.983**
SAM	1.043	0.163	0.416	0.100	0.079	0.077	**0.059**	0.110	0.117	0.075
						σ=70				
PSNR	11.22	34.43	26.78	31.80	37.18	34.96	39.03	38.95	39.16	**39.53**
SSIM	0.019	0.866	0.443	0.865	0.918	0.862	0.949	0.952	0.958	**0.975**
SAM	1.150	0.198	0.521	0.107	0.097	0.088	**0.068**	0.145	0.115	0.085
						blind				
PSNR	14.76	36.64	29.74	31.95	39.16	39.86	41.11	40.92	40.66	**41.28**
SSIM	0.047	0.907	0.579	0.868	0.943	0.951	0.965	0.975	0.971	**0.983**
SAM	1.018	0.158	0.407	0.097	0.077	0.078	**0.057**	0.109	0.116	0.073
Time(s)	-	6510.05	346.37	6538.30	1369.82	49529.20	10946.52	251.67	238.23	**209.15**

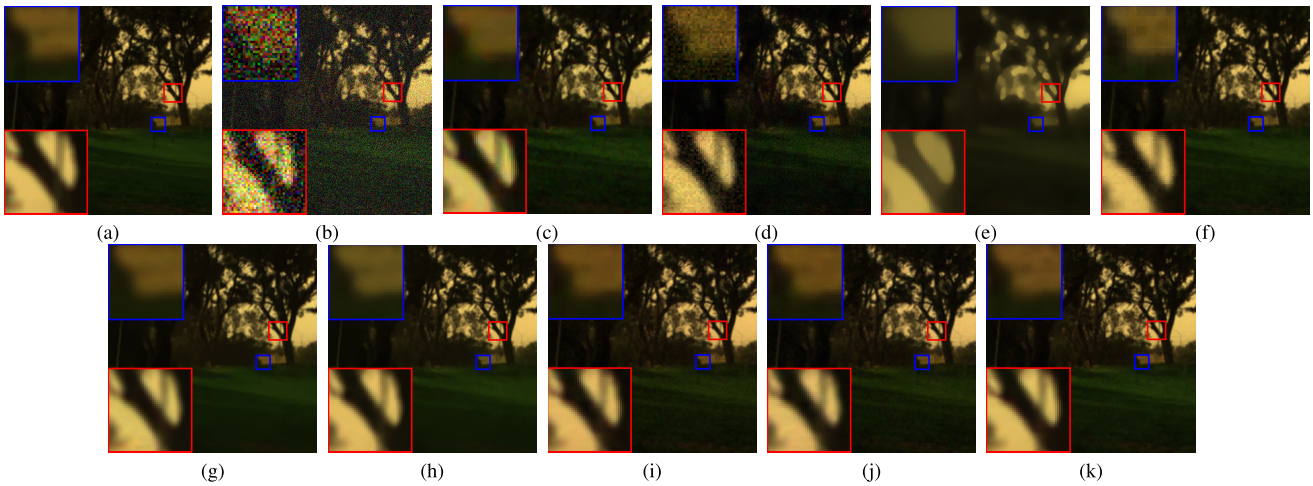

Fig. 7. Visual comparison on a denoised image from the ICVL data set in the Gaussian noise case. Zoomed-in view gives better visualization (a) Clean, (b) Noisy, (c) BM4D, (d) LRMR, (e) LRTV, (f) TDL, (g) LLRT, (h) NGmeet, (i) HSI-DeNet, (j) HSID-CNN, (k) Ours.

TABLE III

PERFORMANCE COMPARISON OF ALL THE COMPETING METHODS IN THE COMPLEX NOISE CASE ON THE ICVL DATA SET. THE BEST AND THE SECOND BEST RESULTS ARE BOLDFACED AND UNDERLINED, RESPECTIVELY

	Noisy HSI	LRMR [17]	LRTV [11]	TDL [14]	NGMeet [26]	NMoG [18]	E3DTV [41]	LRTDTV [22]	HSI-DeNet [27]	HSID-CNN [28]	Ours
						case 1					
PSNR	17.90	29.05	31.94	28.55	29.19	33.64	33.12	39.15	39.57	39.60	**40.28**
SSIM	0.145	0.529	0.867	0.470	0.572	0.788	0.883	0.948	0.971	0.973	**0.983**
SAM	1.008	0.516	0.094	0.528	0.582	0.098	0.179	0.217	0.110	0.114	**0.076**
						case 2					
PSNR	17.65	29.04	31.94	27.56	28.97	34.60	33.18	38.70	39.42	39.61	**40.58**
SSIM	0.138	0.530	0.867	0.442	0.566	0.807	0.885	0.945	0.970	0.973	**0.984**
SAM	1.007	0.515	0.100	0.538	0.584	0.098	0.296	0.213	0.115	0.117	**0.073**
						case 3					
PSNR	17.60	28.37	31.47	26.67	28.29	34.26	32.65	37.30	38.76	38.64	**40.24**
SSIM	0.139	0.516	0.867	0.431	0.564	0.821	0.885	0.933	0.974	0.972	**0.984**
SAM	1.024	0.535	0.119	0.569	0.605	0.117	0.233	0.212	0.114	0.119	**0.075**
						case 4					
PSNR	14.79	24.84	26.36	21.17	24.88	28.69	32.34	36.38	35.56	35.82	**36.54**
SSIM	0.103	0.426	0.733	0.260	0.501	0.624	0.876	0.937	0.921	0.925	**0.938**
SAM	0.997	0.650	**0.117**	0.775	0.739	0.561	0.575	0.227	0.238	0.221	0.192
						case 5					
PSNR	13.86	24.10	25.82	19.87	23.94	27.43	31.60	35.59	34.11	34.20	**35.95**
SSIM	0.084	0.416	0.732	0.231	0.477	0.604	0.875	0.917	0.913	0.910	**0.937**
SAM	1.012	0.659	**0.147**	0.776	0.755	0.573	0.590	0.217	0.215	0.196	0.186
Time(s)	-	374.79	3992.50	2004.88	11727.54	7317.45	13868.21	9810.8	242.75	234.57	**206.25**

than LRTDTV. This is because impulse noise is very difficult to remove, and much less mixture of (non-i.i.d.) Gaussian and impulse noise, and a mixture of all the noise from case 1 to case 4. However, the LRTDTV method has been verified to be very effective in dealing with complex noise and thus obtains excellent performance. In practice, if we want to further

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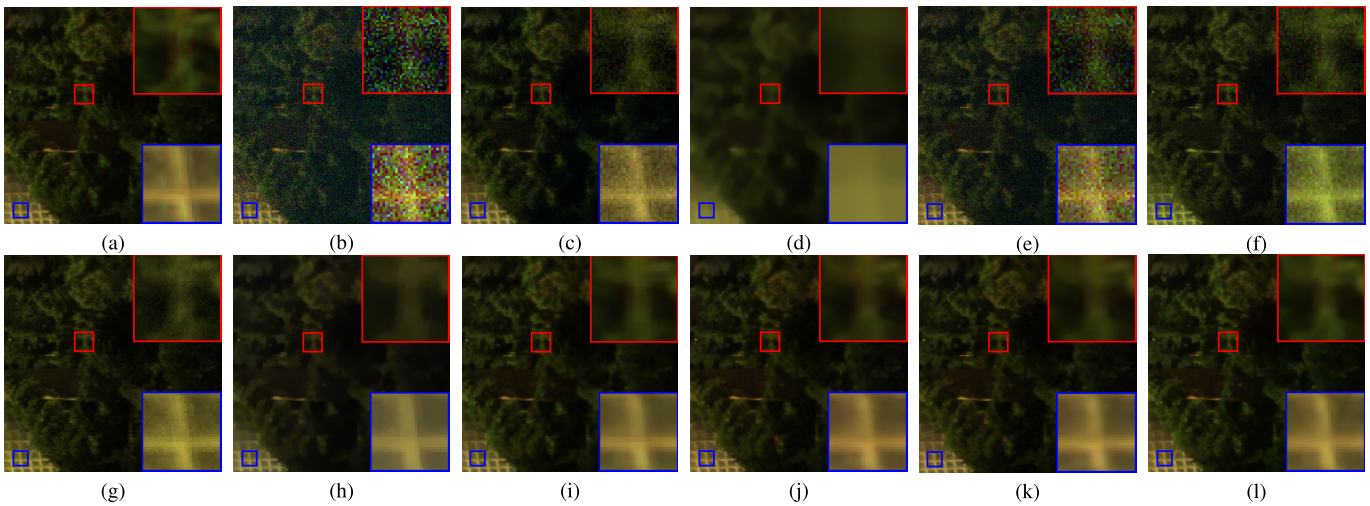

Fig. 8. Visual comparison on a denoised image from the ICVL data set in the complex noise case 1. Zoomed-in view gives better visualization (a) Clean, (b) Noisy, (c) LRMR, (d) LRTV, (e) TDL, (f) NGMeet, (g) NMoG, (h) E3DTV, (i) LRTDTV, (j) HSI-DeNet, (k) HSID-CNN, (l) Ours.

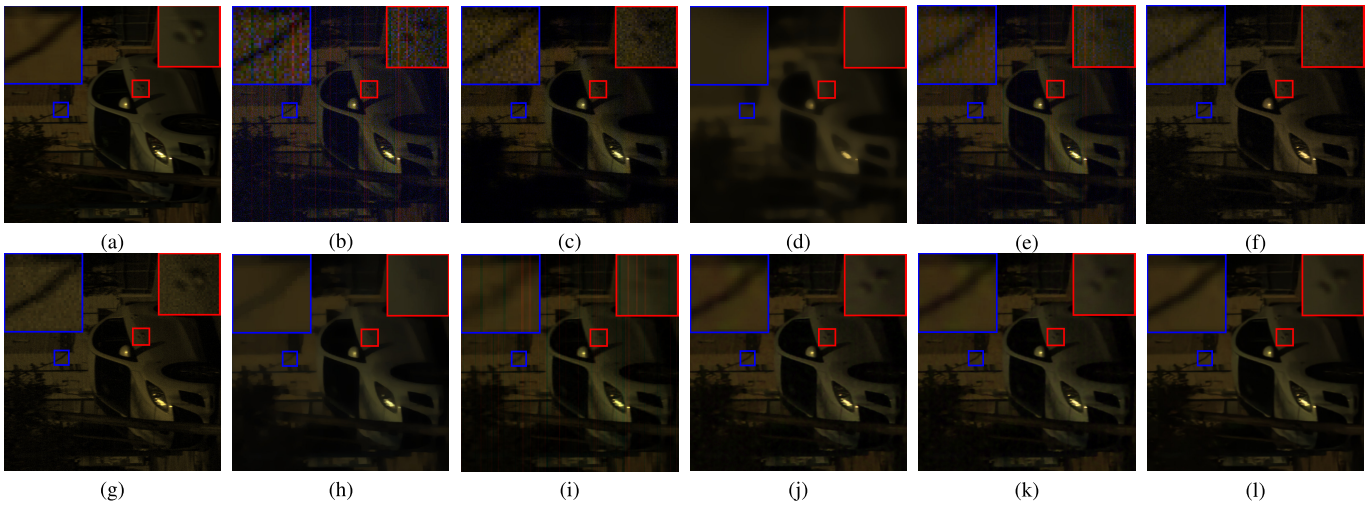

Fig. 9. Visual comparison on a denoised image from the ICVL data set in the complex noise case 2. Zoomed-in view gives better visualization (a) Clean, (b) Noisy, (c) LRMR, (d) LRTV, (e) TDL, (f) NGMeet, (g) NMoG, (h) E3DTV, (i) LRTDTV, (j) HSI-DeNet, (k) HSID-CNN, (l) Ours.

improve the denoising performance of our method as well as other deep learning-based methods in the complex noise cases, e.g., case 4 and case 5, one possible way is to construct more training pairs containing these type of noise and then use these training data to train or finetune the deep learning-based methods. Also, compared with the two deep learning-based methods (HSI-DeNet and HSID-CNN), our method can obtain better denoising performance. Also, the running time of our method is far less than other competing methods.

Furthermore, from Figs. 8–10, we can observe that the competing methods LRMR and LRTV, can hardly remove the complex mixture noise from the images. Also, it can be seen that although NMoG, E3DTV, LRTDTV, HSI-DeNet, and HSID-CNN, can obtain better denoising results, the denoised images still contain some noise or the structure information is not well preserved. Comparatively, our method is able to both finely remove the complex noises and better preserve the structure and details, and thus obtains better visual reconstruction results. Additionally, we show the PSNR values of each band for the three figures in Fig. 11(b)–(d), from which we can observe that our method can obtain higher PSNR values across almost all the bands in comparison with other competing methods.

To further examine the learning capacity of our proposed network, we also test our network on other source HSI data set, such as CAVE data set. We randomly select ten images as testing data. Each image is with size of $512 \times 512 \times 31$. For each image, we add five kinds of complex noise (case 1 to case 5) for testing, and the competing methods are the same as the first synthetic experiment. The final experimental results are shown in Table IV. From Table IV, it can be observed that our method achieves comparable deniosing results compared with other state-of-the-art methods, which further verifies that our proposed network has a certain generalization ability.

B. Real HSI Denoising

In this section, we evaluate the performance of the proposed method on two real HSI data set, the Indian Pines data set[3] and

[3]https://engineering.purdue.edu/ biehl/MultiSpec/hyperspectral.html

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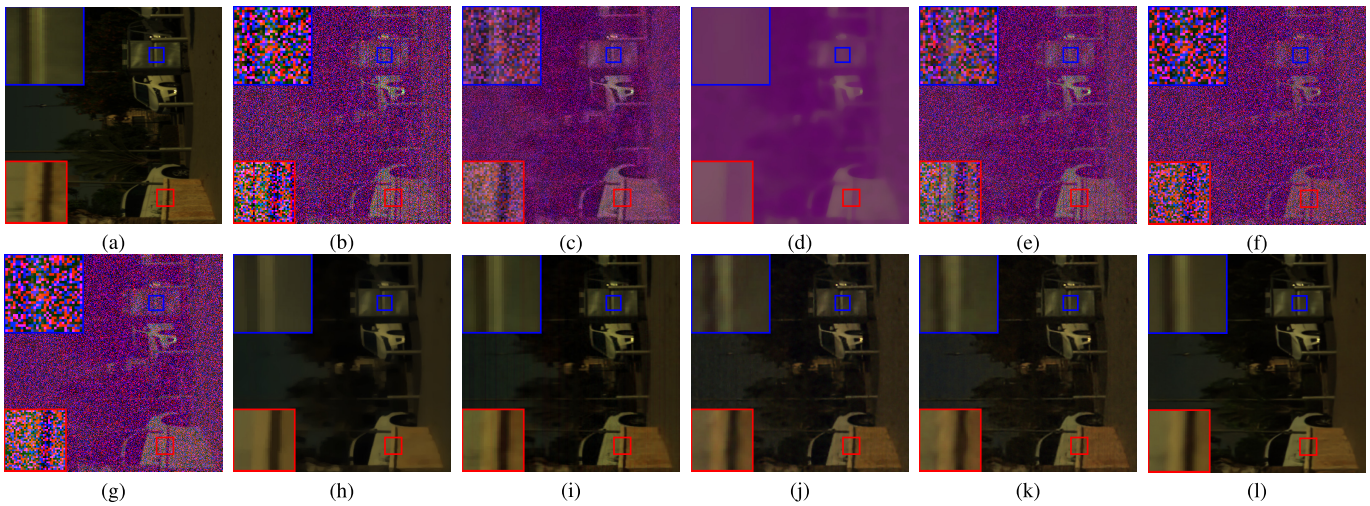

Fig. 10. Visual comparison on a denoised image from the ICVL data set in the complex noise case 5. Zoomed-in view gives better visualization (a) Clean, (b) Noisy, (c) LRMR, (d) LRTV, (e) TDL, (f) NGMeet, (g) NMoG, (h) E3DTV, (i) LRTDTV, (j) HSI-DeNet, (k) HSID-CNN, (l) Ours.

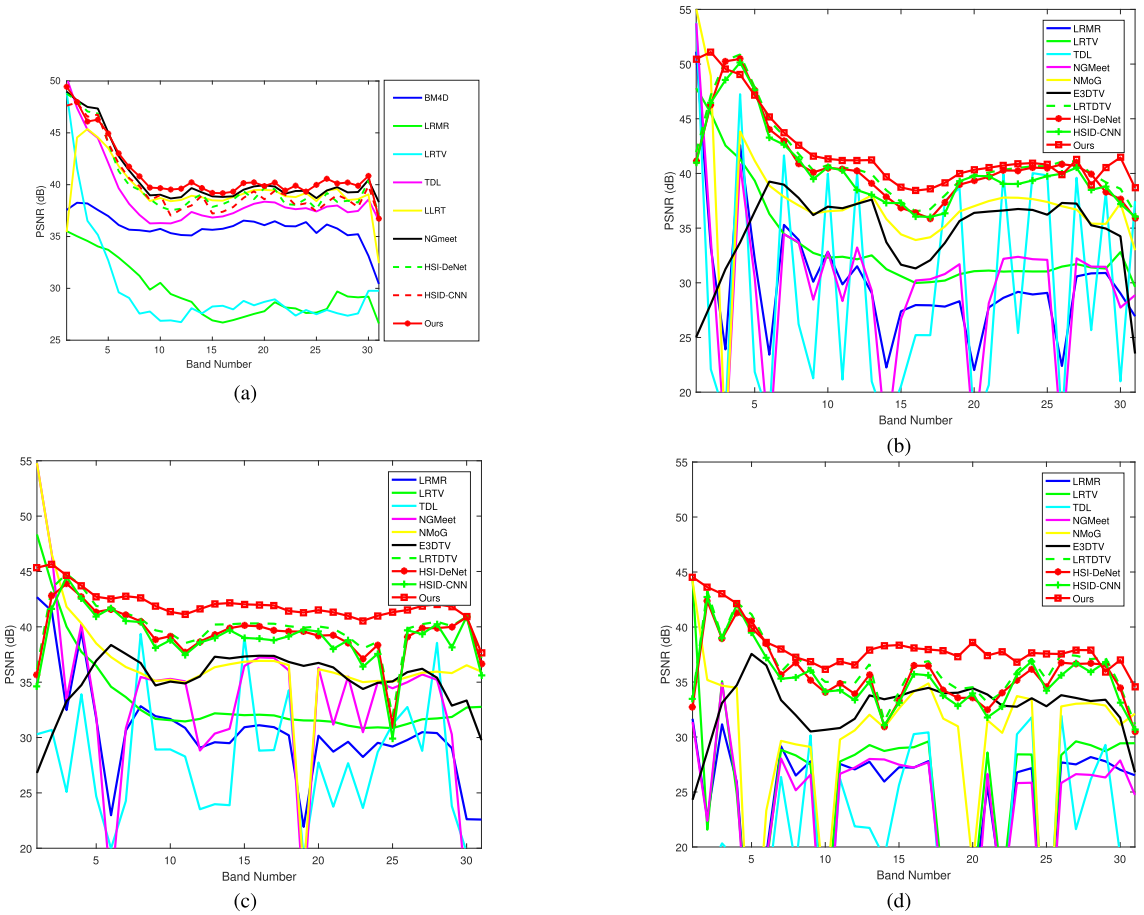

Fig. 11. PSNR values of each band for the images shown from Figs. 7–10 (a) Gaussian Case (σ = 50), (b) Case 1, (c) Case 2, (d) Case 5.

the Urban data set.[4] The first data set is Indian Pines, which contains 220 bands, and each band has the size of 145 × 145. The second one is Urban, which is with size of 307 × 307 × 210. For these two data sets, some bands are seriously polluted by the atmosphere and water and corrupted by complex noises (e.g., deadline, stripe, sparse and Gaussian noise), thus it is a big challenge to remove this noise. Additionally, we use the trained network of the aforementioned synthetic experiment as the pretrained model and then fine-tune this network using a small number of clean bands with additive Gaussian noise. More specifically, for the Indian pines data set, we select the first 31 bands and then add Gaussian noise with noise level

[4]http://www.tec.army.mil/hypercube

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TABLE IV

PERFORMANCE COMPARISON OF ALL THE COMPETING METHODS IN THE COMPLEX NOISE CASES ON THE CAVE DATA SET. THE BEST AND THE SECOND BEST RESULTS ARE BOLDFACED AND UNDERLINED, RESPECTIVELY

	Noisy HSI	LRMR [17]	LRTV [11]	TDL [14]	NGMeet [26]	NMoG [18]	E3DTV [41]	LRTDTV [22]	HSI-DeNet [27]	HSID-CNN [28]	Ours
						case 1					
PSNR	17.47	28.90	28.44	27.81	25.97	31.82	31.20	**34.89**	33.65	34.06	34.47
SSIM	0.153	0.562	0.786	0.469	0.479	0.748	0.888	0.913	0.952	0.957	**0.963**
SAM	1.160	0.759	0.372	0.701	0.796	0.549	0.380	0.319	0.258	0.247	**0.230**
						case 2					
PSNR	17.72	29.09	28.54	26.62	26.37	31.63	31.25	34.40	33.82	34.11	**34.53**
SSIM	0.155	0.570	0.789	0.433	0.504	0.735	0.889	0.915	0.950	0.956	**0.964**
SAM	1.153	0.755	0.360	0.741	0.796	0.542	0.382	0.323	0.255	0.248	**0.231**
						case 3					
PSNR	17.26	28.30	28.01	25.95	25.68	31.31	30.84	34.07	33.56	33.98	**34.16**
SSIM	0.149	0.557	0.787	0.437	0.487	0.742	0.895	0.908	0.948	0.957	**0.962**
SAM	1.169	0.776	0.384	0.739	0.835	0.532	0.371	0.321	0.253	0.251	**0.231**
						case 4					
PSNR	14.98	24.42	23.46	21.05	22.59	26.34	30.64	34.45	32.57	33.08	**34.69**
SSIM	0.122	0.483	0.635	0.281	0.488	0.601	0.877	0.904	0.925	0.931	**0.946**
SAM	1.112	0.821	0.763	0.916	0.908	0.769	0.385	0.338	0.277	0.269	**0.248**
						case 5					
PSNR	14.29	24.05	23.33	20.68	22.28	25.14	30.42	33.65	32.30	33.36	**34.24**
SSIM	0.102	0.481	0.639	0.254	0.456	0.592	0.889	0.901	0.918	0.935	**0.943**
SAM	1.127	0.841	0.779	0.929	0.928	0.825	0.374	0.331	0.293	0.302	**0.270**
Time(s)	-	75.51	1555.65	660.47	3986.23	3621.26	2192.24	871.67	49.08	48.25	**46.80**

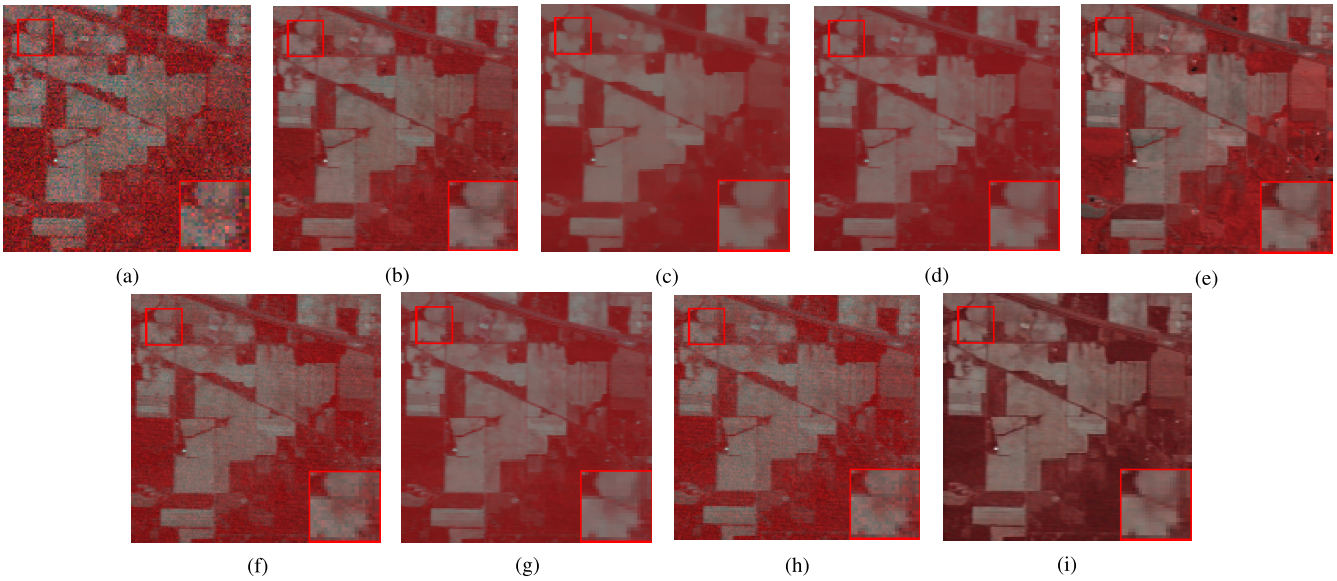

Fig. 12. False color images of the denoised results on band 160, 215 and 217 of the indian pines data set for all the competing methods. Zoomed-in view gives better visualization (a) Noisy, (b) LRMR, (c) LRTV, (d) LRTDTV, (e) NMoG, (f) E3DTV, (g) HSI-DeNet, (h) HSID-CNN, (i) Ours.

$\sigma = 15$ to these bands. Then, this training pair is used for finetuning our network. As for the urban data set, we also choose the first 31 bands and use a similar way to construct the training pair for finetuning. For the testing of the two data sets, we split the remaining bands into a series of sub-data sets, of which each contains 31 bands.

In this experiment, the competing methods include LRMR [17], LRTV [11], E3DTV [41], NMoG [18], LRTDTV [22], and two deep learning methods (HSI-DeNet [27], HSID-CNN [28]). Additionally, to comprehensively compare the denoising effect, we show some false color images of the reconstructed results of the Indian Pines and Urban data sets from Figs. 12–15. For easy observation, an area of interest is amplified in the restored images.

Specifically, Figs. 12 and 13 show the recovered HSIs of the Indian pines data set by all the compared methods. From Fig. 12, it can be easily seen that other competing methods can not completely remove the complex noise, while our proposed method can almost remove most of the complex noise. From Fig. 13, we can observe that our method can achieve better denoising performance than most of the competing methods, and obtain comparable performance with the NMoG method. Additionally, Figs. 14 and 15 show the recovered HSIs of the urban data sets by the competing methods. From Fig. 14, it can be observed that our method can obtain better and comparable denoising performance compared with other methods. From Fig. 15, we can observe that all these methods except the NMoG, cannot completely remove the horizontal stripe noise,

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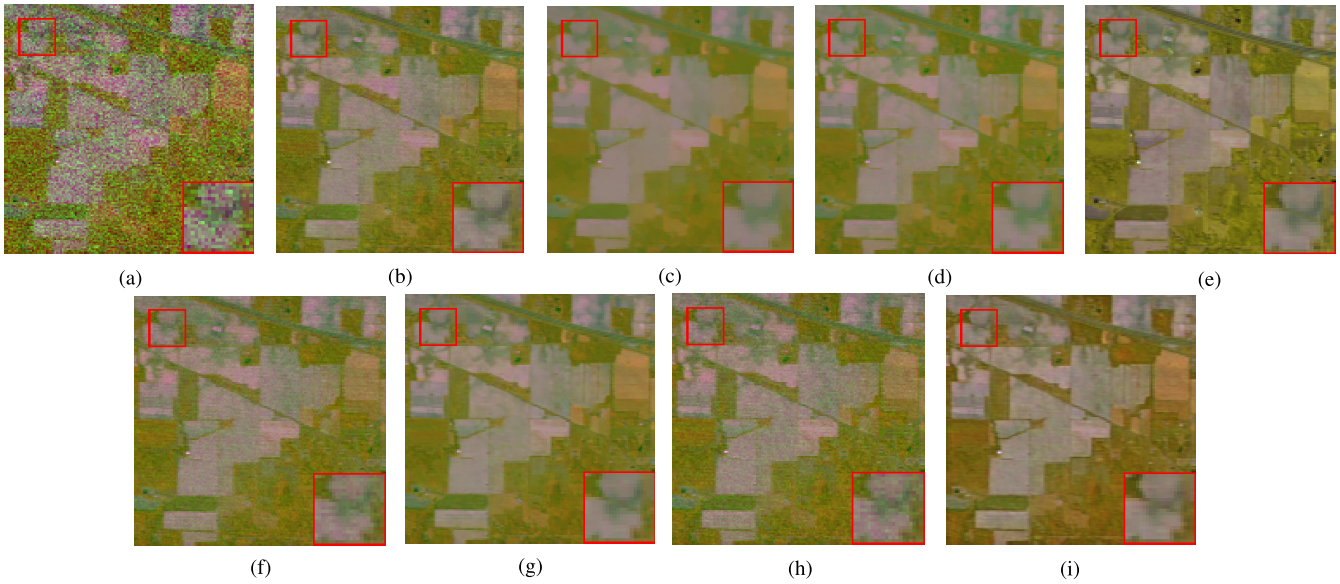

Fig. 13. False color images of the denoised results on band 100, 150 and 200 of the Indian pines data set for all the competing methods. Zoomed-in view gives better visualization (a) Noisy, (b) LRMR, (c) LRTV, (d) LRTDTV, (e) NMoG, (f) E3DTV, (g) HSI-DeNet, (h) HSID-CNN, (i) Ours.

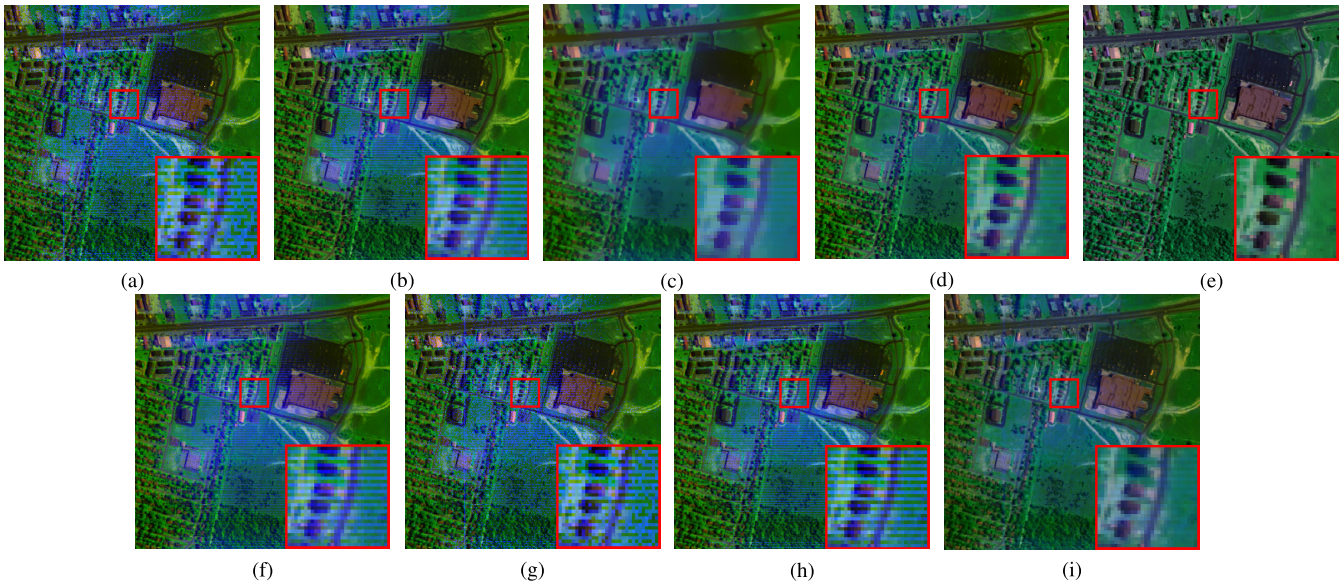

Fig. 14. False color images of the denoised results on band 50, 100 and 150 of the urban data set for all the competing methods. Zoomed-in view gives better visualization (a) Noisy, (b) LRMR, (c) LRTV, (d) LRTDTV, (e) NMoG, (f) E3DTV, (g) HSI-DeNet, (h) HSID-CNN, (i) Ours.

while our method performs better in both detail preservation and noise removing.

C. Ablation Study

As for our proposed method, the core modules are the two proposed global reasoning modules, namely GSM and GCM. To illustrate their necessity and effectiveness, we first conduct the ablation study on the effect of the proposed GSM and GCM modules by adding them to the baseline network, which contains only the DCMs. Specifically, this experiment is conducted over the ICVL data set, and we adopt the same training strategy as the synthetic experiment. The testing experiment is conducted only on the blind case. The results are reported in Table V.

From Table V, we can observe that the baseline model only containing DCMs, obtains the PSNR=38.64 and SSIM=0.958, which has outperformed most competing methods due to the extracted rich local features but inferior to TDL, LLRT NGMeet, and HSI-DeNet. Then, by adding the GSM to the baseline model, we can achieve an improvement of 1.21 and 0.011 on average PSNR and SSIM, respectively. This is because the GSM is helpful to remove spatially relevant noise and preserve image structures. Similarly, by adding the GCM to the baseline model, we can improve the baseline by 1.28 and 0.013 on average PSNR and SSIM, respectively. This is because the GCM can help capture the correlations between channels of the feature map, and thus can utilize more global information. Furthermore, by adding the two global reasoning

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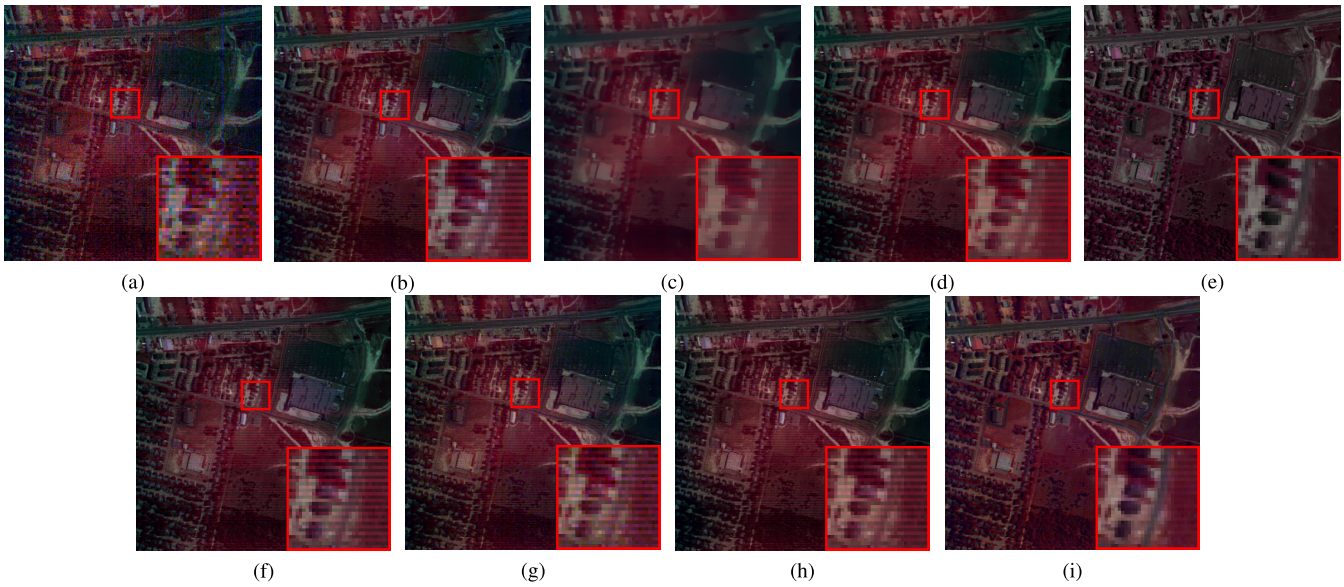

Fig. 15. False color images of the denoised results on band 108, 200 and 207 of the urban data set for all the competing methods. Zoomed-in view gives better visualization (a) Noisy, (b) LRMR, (c) LRTV, (d) LRTDTV, (e) NMoG, (f) E3DTV, (g) HSI-DeNet, (h) HSID-CNN, (i) Ours.

TABLE V
ABLATION STUDY ON THE EFFECT OF THE PROPOSED GSM AND GCM MODULES BY ADDING THEM TO THE BASELINE NETWORK

Models	PSNR	SSIM
Baseline	38.64	0.958
Baseline + GSM	39.85	0.969
Baseline + GCM	39.92	0.971
Baseline + GSM + GCM	41.28	0.983

TABLE VI
ABLATION STUDY ON THE EFFECT OF THE PROPOSED GSM AND GCM MODULES BY ELLIMINATING THEM FROM THE PROPOSED NETWORK

Models	PSNR	SSIM
Proposed network	41.28	0.983
Proposed network w/o GSM	40.15	0.965
Proposed network w/o GCM	40.39	0.970

modules (namely GSM and GCM) to the baseline model, we can obtain the best result, which significantly increases the average PSNR and SSIM by 2.64 and 0.025, respectively. This is because the decoder can receive the two orthogonal contextual information, and thus it is able to extract and utilize more comprehensive representations for the HSI denoising.

Furthermore, we have conducted one additional experiment to demonstrate the effectiveness of GSM and GCM by eliminating them from the proposed network. Specifically, we use the final proposed graph reasoning network as the baseline network and then obtain two variants of this network by eliminating the GSM term and GCM term from the proposed network, respectively. The training and test stages are the same as the above experiment. The final performance changes in terms of PSNR and SSIM are illustrated in Table VI.

From Table VI, it can be easily seen that by eliminating the GSM from the proposed network, the average PSNR and SSIM decrease by 1.13 and 0.018, respectively. This is because GSM can capture the global spatial information and thus verifies that the global spatial information is really helpful to improve the denoising results. Conversely, the denoising results decline without this module as is shown in Table VI. Similarly, by eliminating the GCM from the proposed network, the average PSNR and SSIM decrease by 0.89 and 0.013, respectively. This is because GCM can capture the global channel information and thus proves the global channel information is really beneficial to improve the denoising results. Conversely, the denoising results decline without this module as is shown in Table VI.

IV. CONCLUSION AND FUTURE WORK

In this article, we have introduced a deep spatial-spectral global reasoning network for HSI denoising. To effectively and efficiently explore the contextual information for HSI denoising, we proposed two global reasoning modules to capture context along the spatial and channel dimensions, respectively. These two modules are further combined with densely connected networks and form the encoder and decoder, which can thus be used to exploit rich feature representations. The network formed by the encoder and decoder can preserve the image structures well while removing the complex noise. Experiments have verified the superiority of our method on both synthetic and real HSI data sets compared with other state-of-the-art HSI denoising methods.

Although the proposed network achieves excellent performance, the design of the network still lacks interpretation. In the future, we may focus on designing an interpretable network deduced from the traditional HSI denoising method (e.g., unfolding an iterative algorithm into a network), and thus each module of the proposed network has its own physical

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interpretation. Additionally, by embedding these traditional prior into the design of the network, we may achieve better performance in the same training data sets compared with other deep learning-based methods.

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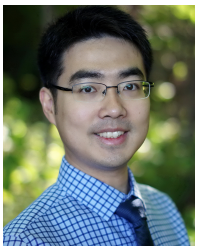

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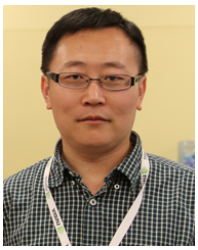

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